Ex:No: 1I Date: 04.02.26	Car Inventory Management Using Data Frame and Factor

Problem Description:

A car dealership wants to maintain an **inventory of cars**, including:

- Car Model (string)
- Registration Number (string)
- Price (numeric)
- Availability (logical: TRUE/FALSE)
- Car Type (categorical: Sedan, SUV, Hatchback)

The dealership wants to:

1. Store car data in a **data frame**.
2. Represent **Car Type as a factor**.
3. **Search cars** based on **availability** and **type**.

All data should be **entered by the user dynamically**.

Objective:

To use data frames and factors in R to store heterogeneous car inventory data, categorize car types, and search for cars based on availability and type.

Sample Input
Enter number of cars: 3 Enter details for Car 1 : Enter car model: Volkswagen Enter registration number: V123 Enter price: 2000000 Is the car available? (TRUE/FALSE): TRUE Enter car type (Sedan/SUV/Hatchback): Sedan Enter details for Car 2 : Enter car model: Ford Enter registration number: F123 Enter price: 1500000 Is the car available? (TRUE/FALSE): TRUE Enter car type (Sedan/SUV/Hatchback): SUV Enter details for Car 3 : Enter car model: BMW Enter registration number: B123 Enter price: 3000000 Is the car available? (TRUE/FALSE): FALSE Enter car type (Sedan/SUV/Hatchback): Hatchback

Expected Output

```
Complete Car Inventory:
       Model RegNo   Price Available      Type
1 Volkswagen  V123 2000000      TRUE     Sedan
2       Ford  F123 1500000      TRUE       SUV
3        BMW  B123 3000000     FALSE Hatchback

Do you want to see available cars only? (TRUE/FALSE): TRUE

Available Cars:
       Model RegNo   Price Available  Type
1 Volkswagen  V123 2000000      TRUE Sedan
2       Ford  F123 1500000      TRUE   SUV

Enter car type to search (Sedan/SUV/Hatchback): SUV

Cars of type SUV :
  Model RegNo   Price Available Type
2  Ford  F123 1500000      TRUE  SUV
```

Program:

User input num_cars

```
num_cars <- as.integer(readline("Enter number of cars:"))
```

Initialize empty vectors Models, RegNos, Prices, Available, CarType

```
models <- c()
RegNos <- c()
Prices <- c()
Available <- c()
CarType <- c()
```

Input car details

```
for(i in 1:num_cars){
  cat("\n Enter details for car", i, ":\n")
  model <- readline("Enter car model: ")
  regno <- readline("Enter registration number: ")
  price <- as.numeric(readline("Enter price: "))
  avail <- as.logical(readline("\n Is the car available?
                                (TRUE/FALSE): "))
```

INPUT:

Enter number of cars: 1

Enter details for car 1:

Enter car model: Alto

Enter registration number: A101

Enter price: 400000

Is the car available? (TRUE / FALSE): TRUE

Enter car type (Sedan / SUV / Hatchback): Hatchback

OUTPUT:

Complete car Inventory:

Model	RegNo	Price	Available	Type
Alto	A101	4000000	TRUE	Hatchback.

Do you want to see available cars only? (TRUE / FALSE): TRUE

Available cars:

	Model	RegNo	Price	Available	Type
1	Alto	A101	400000	TRUE	Hatchback

Enter car type to search (Sedan / SUV / Hatchback): Hatchback

Cars of type Hatchback:

Model	RegNo	Price	Available	Type
Alto	A101	400000	TRUE	Hatchback

```
type <- readline("Enter car type (sedan|SUV|Hatchback): ")
Models <- c(models, model)
Reg Nos <- c(Reg Nos, regno)
prices <- c(prices, price)
Available <- c(Available, avail)
Cartype <- c(Cartype, type)
}
```

Convert CarType to factor

```
carType <- factor(carType, levels = c("Sedan", "SUV", "Hatchback"))
```

Create data frame car_df

```
car_df <- data.frame(
  Model = models,
  RegNo = Reg Nos,
  Price = Prices,
  Available = Available,
  Type = CarType
)
```

Display complete car inventory

```
cat("\n Complete Car Inventory :\n")
print(car_df)
```

Search cars by availability

```
choice <- as.logical (readline ("\n Do you want to see available
                cars only? (TRUE/FALSE): "))
if (choice == TRUE){
    available_cars <- car_df [car_df $ Available == TRUE]
cat ("\n Available cars: \n")
print (available_cars)
}
```

Search cars by type

```
search_type <- readline ("Enter car type to search (Sedan/
                SUV/Hatchback): ")
type_cars <- car_df [car_df $ Type == search_type]
cat ("\n Cars of type ", search_type ,": \n")
print(type_cars)
```

Result:

Thus, the program successfully demonstrates the use of **data frames and factors in R** to store **heterogeneous car inventory data**, categorize cars by **type**, and **search cars based on availability and type.**